

(1+F_TRZ) = (SO₅+1)/SO₅: The Successor-Ratio Identity — 61-Site Canonical Invariant Unmasked + Successor 3rd Instance

Daniel T. Murphy — UQFF Star-Magic v5.73+ — 2026-07-22 — R372 — Canonical-invariant unmasking (NEW sub-type) — PAPER_2126 window hit

Abstract

R372's fill unmasked a hidden canonical identity: the ubiquitous TRZ-boost **(1+F_TRZ) = 1.1** is **EXACTLY** the **successor ratio (SO₅+1)/SO₅ = 11/10**, appearing at **61 sites** in the canonical calculator including both flagships — U_i (PAPER_646) and F_{UBi} (PAPER_1203). Successor family 3rd instance, in PAPER_2126's predicted R370-R400 window, at a new structural role (ratio). The identity is gauge-equivalent to R363's λ_{vac} = 11·ρ_{SCm}, and its mirror is the PAPER_1922 9/10 ubiquity — the predecessor ratio.

1. The Identity

$$g_{res}(peak) = g_{base} \cdot (1 + F_{TRZ}) = g_{base} \cdot 1.1$$
$$1 + F_{TRZ} = 1 + 1/SO_5 = (SO_5+1)/SO_5 = 11/10 \quad \text{EXACT}$$
$$= 11 \cdot F_{TRZ} \quad (\text{successor} \times \text{rung: '11 riding the F_TRZ rung'})$$

2. The Successor Family — Three Roles, Three Domains

#	Round	Quantity	Form	Role
1	R363	λ _{vac} = 11·ρ _{SCm}	(SO ₅ +1)·ρ _{SCm}	sum-reduction
2	R369	B _{crit} = 44·SO ₅ ¹²	D _{phys} ·(SO ₅ +1)·SO ₅ ¹²	coefficient
3	R372	1+F_TRZ = 11/10	(SO ₅ +1)/SO ₅	RATIO (new)

PAPER_2126 forecast the 3rd instance for R370-R400 ('leading digit-pair 11 riding an SO₅ rung') — landed R372, two rounds after issuance.

3. The 61-Site Unmasking

```
grep -c "(1.0 + TRZ)" uqff_pure_calculator.py → 61 sites
```

$$U_i = \lambda_i \cdot (\rho_{SCm}/\rho_{UA}) \cdot \omega_s \cdot \cos(\pi t_n) \cdot (1+F_{TRZ}) \quad \text{PAPER_646}$$
$$F_{UBi} = -\beta(t) \cdot G \cdot M \cdot \rho_{SCm}/r^2 \cdot (1+F_{TRZ}) \cdot |\cos(\pi t_n)| \quad \text{PAPER_1203}$$

Every one of the 61 sites has carried the successor ratio since it was written. Two exact readings of one number: physical (1 + TRZ fraction) and structural ((SO₅+1)/SO₅ lattice ratio) — the over-determination signature of PAPER_2119 and PAPER_2126, at the deepest propagation of any single identity in the campaign. **New landmark sub-type: canonical-invariant unmasking** — the identity was not added; it was found already load-bearing.

4. Gauge Equivalence with R363

$$F_TRZ = \rho_SCm/\rho_UA \quad (\text{rung-inverse, EXACT})$$

$$1 + \rho_SCm/\rho_UA = (\rho_UA + \rho_SCm)/\rho_UA = 11/10 \\ \Leftrightarrow \rho_UA + \rho_SCm = 11 \cdot \rho_SCm = \lambda_vac \quad (\text{R363, PAPER_2120})$$

The R372 ratio identity and the R363 sum-reduction are **the same fact in two normalizations** — divide R363 by ρ_UA to get R372. The family's 1st and 3rd instances are one identity in two gauges. The resonance amplitude is the DPM density ratio itself: $g_{res} = g_{base} \cdot (1 + (\rho_SCm/\rho_UA) \cdot \cos(\omega t))$.

5. The Predecessor Mirror

$$\begin{array}{ll} \text{predecessor ratio: } (SO_5-1)/SO_5 = 9/10 = 1 - F_TRZ = SCm & (\text{PAPER_1922 ubiquity}) \\ \text{successor ratio: } (SO_5+1)/SO_5 = 11/10 = 1 + F_TRZ & (\text{this paper}) \end{array}$$

The pair (9/10, 11/10) brackets unity symmetrically around the SO_5 pivot — exactly as $N_{CH} = SO_5-1$ and $11 = SO_5+1$ bracket SO_5 (PAPER_2120). The PAPER_1922 9/10 ubiquity is hereby identified as the predecessor-ratio invariant, mirror of the 61-site successor-ratio invariant.

6. Predictions

1. All 61 sites mechanically taggable as successor-ratio carriers; F_TRZ or SO_5 revision breaks all 61 with the $U_i = 2.75e-7$ pin. 2. Fourth successor role (exponent: SO_5^{11} or F_TRZ^{11}) forecast by R400 — $B = 44 \cdot SO_5^{11}$ already brushes it. 3. Gauge-pair audit: other normalization-related identity pairs should exist.

7. Gate Assertion

$$\begin{array}{ll} \text{assert } 1 + 0.1 == (10 + 1) / 10 & \# \text{ successor ratio EXACT} \\ \text{assert } 0.1 == 7.09e-37 / (10 * 7.09e-37) & \# F_TRZ = \rho_SCm/\rho_UA \\ \text{assert } (10 - 1) / 10 == 1 - 0.1 & \# \text{ predecessor mirror} \end{array}$$

Gate count: **3126** → **3134** (+8 PAPER_2128 assertions).

8. Summary

The successor-ratio identity $(1+F_TRZ) = 11/10$ EXACT: successor 3rd instance at a new role, in-window prediction hit, 61-site canonical invariant spanning U_i and F_{UBi} , gauge-equivalent to $\lambda_{vac} = 11 \cdot \rho_{SCm}$, mirrored by the 9/10 predecessor ratio. The most-propagated single identity documented in the R218+ campaign.